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सत्यमेव जयते
GOVERNMENT OF INDIA
MINISTRY OF SKILL DEVELOPMENT
& ENTREPRENEURSHIP



NCVT
कौशल गुणवत्ता प्रगति



NSDC
REIMAGINE FUTURE

Qualification Pack



Software Programmer

Electives: Java/ Python/ HTML 5

QP Code: SSC/Q0510

Version: 1.0

NSQF Level: 4

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Qualification Pack

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SSC/Q0510: Software Programmer

Brief Job Description

This unit is about contributing to software programming where both the business impact and technical complexity are low

Personal Attributes

They must show enthusiasm to learn new things, they must have good communication skills to communicate with colleagues and stake holders

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SSC/N0509: Verify the specifications and contribute to the design of software program](#)
2. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Electives(mandatory to select at least one):

Elective 1: Java

This elective deals with Programming in Java

1. [SSC/N0510: Develop, test and execute software programs as per specifications using Java](#)

Elective 2: Python

This electives deal with Programming in Python

1. [SSC/N0511: Develop, test and execute software programs as per specifications using Python](#)

Elective 3: HTML 5

This electives deals with Programming in HTML 5

1. [SSC/N0512: Write, test and execute programs in HTML5 to develop web or mobile applications as per specifications](#)

Qualification Pack (QP) Parameters

Qualification Pack

Sector	IT-ITeS
Sub-Sector	IT Services
Occupation	Application Deployment
Country	India
NSQF Level	4
Credits	21
Aligned to NCO/ISCO/ISIC Code	NCO-2015/2512.0100
Minimum Educational Qualification & Experience	11th Class OR 10th Class (1-Yr NTC/NAC/CITS) OR Certificate-NSQF (Level 3 STT)
Minimum Level of Education for Training in School	10th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	16 Years
Last Reviewed On	NA
Next Review Date	17/11/2025
NSQC Approval Date	17/11/2022
Version	1.0
Reference code on NQR	2022/ITES/ITSSC/06478
NQR Version	1

Qualification Pack

SSC/N0509: Verify the specifications and contribute to the design of software program

Description

This NOS is about understanding the business and technical requirements of the software program to be developed

Scope

The scope covers the following :

- Requirement Analysis, Design Analysis, Best Practices

Elements and Performance Criteria

Requirement Analysis

To be competent, the user/individual on the job must be able to:

PC1. Understand the business context and business/ end-user requirements for the program

PC2. Understand functional and non-functional requirements for the program

Design Analysis

To be competent, the user/individual on the job must be able to:

PC3. Interpret the high-level and low-level design parameters using appropriate sources

PC4. Participate in the design of software program

Best practices

To be competent, the user/individual on the job must be able to:

PC5. Understand the project management frameworks used (agile methodologies and design thinking principles) in planning the software program

PC6. Keep a track of latest technological developments related to programming and identify their potential applications in your business domain

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. organizational policies, procedures and guidelines which relate to SDLC and programming

KU2. different data sources and how to access documents and information from data sources

KU3. organizational policies and procedures for documentation

KU4. who to involve while at various stages of SDLC

KU5. the range of standard templates and tools available and how to use them

KU6. overview of IR 4.0 technologies and their use cases

KU7. basic approaches in design thinking and agile methodologies

KU8. programming constructs and coding practices relevant to the programming language

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- KU9.** tools, platforms and frameworks required for code development in the programming language
- KU10.** basic techniques to secure software code
- KU11.** tools for documentation, data visualization and analysis
- KU12.** fundamentals of requirement analysis

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** seek clarification and advice from appropriate people
- GS2.** listen effectively and orally communicate information accurately
- GS3.** work effectively in a customer facing environment
- GS4.** create accurate and well written documents/reports
- GS5.** understand the impact of business related decisions on organization
- GS6.** follow instructions, guidelines, procedures, rules and service level agreements
- GS7.** follow rule-based decision-making processes
- GS8.** plan and organize work to achieve targets and deadlines

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Requirement Analysis</i>	14	26	-	-
PC1. Understand the business context and business/ end-user requirements for the program	7	13	-	-
PC2. Understand functional and non-functional requirements for the program	7	13	-	-
<i>Design Analysis</i>	14	26	-	-
PC3. Interpret the high-level and low-level design parameters using appropriate sources	7	13	-	-
PC4. Participate in the design of software program	7	13	-	-
<i>Best practices</i>	13	7	-	-
PC5. Understand the project management frameworks used (agile methodologies and design thinking principles) in planning the software program	6	3	-	-
PC6. Keep a track of latest technological developments related to programming and identify their potential applications in your business domain	7	4	-	-
NOS Total	41	59	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N0509
NOS Name	Verify the specifications and contribute to the design of software program
Sector	IT-ITes
Sub-Sector	IT Services
Occupation	Application Deployment
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	NA
Next Review Date	17/11/2025
NSQC Clearance Date	17/11/2022

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DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	31/03/2022
Next Review Date	31/03/2025
NSQF Clearance Date	31/03/2022

Qualification Pack

SSC/N0510: Develop, test and execute software programs as per specifications using Java

Description

This NOS is about program development and execution in java aligned to the prescribed technical specifications

Scope

The scope covers the following :

- Design Verification, Programming and Testing, Documentation

Elements and Performance Criteria

Design Verification

To be competent, the user/individual on the job must be able to:

- PC1.** Check the design and code specifications for the program to be developed
- PC2.** Define the objectives and functionality of the Java program, and align it to design specifications

Programming and Testing

To be competent, the user/individual on the job must be able to:

- PC3.** Develop software code for various applications in Java
- PC4.** Implement coding best practices such as reusability, naming conventions, portability, etc.
- PC5.** Create appropriate test cases and test the Java program
- PC6.** Debug the code and review the code with appropriate stakeholders

Documentation

To be competent, the user/individual on the job must be able to:

- PC7.** Maintain a record of test cases, software program, exception handling, reusable modules, etc

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational policies, procedures and guidelines which relate to SDLC and programming
- KU2.** different data sources and how to access documents and information from data sources
- KU3.** organizational policies and procedures for documentation
- KU4.** who to involve while at various stages of SDLC
- KU5.** the range of standard templates and tools available and how to use them
- KU6.** overview of IR 4.0 technologies and their use cases
- KU7.** basic approaches in design thinking and agile methodologies
- KU8.** programming constructs and coding practices relevant to the programming language

Qualification Pack

- KU9.** tools, platforms and frameworks required for code development in the programming language
- KU10.** basic techniques to secure software code
- KU11.** tools for documentation, data visualization and analysis
- KU12.** fundamentals of requirement analysis

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** seek clarification and advice from appropriate people
- GS2.** listen effectively and orally communicate information accurately
- GS3.** work effectively in a customer facing environment
- GS4.** create accurate and well written documents/reports
- GS5.** understand the impact of business related decisions on organization
- GS6.** follow instructions, guidelines, procedures, rules and service level agreements
- GS7.** follow rule-based decision-making processes
- GS8.** plan and organize work to achieve targets and deadlines

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Design Verification</i>	8	14	-	-
PC1. Check the design and code specifications for the program to be developed	4	6	-	-
PC2. Define the objectives and functionality of the Java program, and align it to design specifications	4	8	-	-
<i>Programming and Testing</i>	25	47	-	-
PC3. Develop software code for various applications in Java	10	20	-	-
PC4. Implement coding best practices such as reusability, naming conventions, portability, etc.	4	8	-	-
PC5. Create appropriate test cases and test the Java program	6	10	-	-
PC6. Debug the code and review the code with appropriate stakeholders	5	9	-	-
<i>Documentation</i>	2	4	-	-
PC7. Maintain a record of test cases, software program, exception handling, reusable modules, etc	2	4	-	-
NOS Total	35	65	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N0510
NOS Name	Develop, test and execute software programs as per specifications using Java
Sector	IT-ITes
Sub-Sector	IT Services
Occupation	Application Deployment
NSQF Level	4
Credits	31
Version	1.0
Last Reviewed Date	NA
Next Review Date	17/11/2025
NSQC Clearance Date	17/11/2022

Qualification Pack

SSC/N0511: Develop, test and execute software programs as per specifications using Python

Description

This NOS is about program development and execution in Python aligned to the prescribed technical specifications

Scope

The scope covers the following :

- Design Verification, Programming and Testing, Documentation

Elements and Performance Criteria

Design Verification

To be competent, the user/individual on the job must be able to:

- PC1.** Check the design and code specifications for the program to be developed
- PC2.** Define the objectives and functionality of the Python program, and align it to design specifications

Programming and Testing

To be competent, the user/individual on the job must be able to:

- PC3.** Develop software code for various applications in Python
- PC4.** Implement coding best practices such as reusability, naming conventions, portability, etc.
- PC5.** Create appropriate test cases and test the Python program
- PC6.** Debug the code and review the code with appropriate stakeholders

Documentation

To be competent, the user/individual on the job must be able to:

- PC7.** Maintain a record of test cases, software program, exception handling, reusable modules, etc

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational policies, procedures and guidelines which relate to SDLC and programming
- KU2.** different data sources and how to access documents and information from data sources
- KU3.** organizational policies and procedures for documentation
- KU4.** who to involve while at various stages of SDLC
- KU5.** the range of standard templates and tools available and how to use them
- KU6.** overview of IR 4.0 technologies and their use cases
- KU7.** basic approaches in design thinking and agile methodologies
- KU8.** programming constructs and coding practices relevant to the programming language

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- KU9.** tools, platforms and frameworks required for code development in the programming language
- KU10.** basic techniques to secure software code
- KU11.** tools for documentation, data visualization and analysis
- KU12.** fundamentals of requirement analysis

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** seek clarification and advice from appropriate people
- GS2.** listen effectively and orally communicate information accurately
- GS3.** work effectively in a customer facing environment
- GS4.** create accurate and well written documents/reports
- GS5.** understand the impact of business related decisions on organization
- GS6.** follow instructions, guidelines, procedures, rules and service level agreements
- GS7.** follow rule-based decision-making processes
- GS8.** plan and organize work to achieve targets and deadlines

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Design Verification</i>	8	14	-	-
PC1. Check the design and code specifications for the program to be developed	4	6	-	-
PC2. Define the objectives and functionality of the Python program, and align it to design specifications	4	8	-	-
<i>Programming and Testing</i>	25	47	-	-
PC3. Develop software code for various applications in Python	10	20	-	-
PC4. Implement coding best practices such as reusability, naming conventions, portability, etc.	4	8	-	-
PC5. Create appropriate test cases and test the Python program	6	10	-	-
PC6. Debug the code and review the code with appropriate stakeholders	5	9	-	-
<i>Documentation</i>	2	4	-	-
PC7. Maintain a record of test cases, software program, exception handling, reusable modules, etc	2	4	-	-
NOS Total	35	65	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N0511
NOS Name	Develop, test and execute software programs as per specifications using Python
Sector	IT-ITes
Sub-Sector	IT Services
Occupation	Application Deployment
NSQF Level	4
Credits	9
Version	1.0
Last Reviewed Date	NA
Next Review Date	17/11/2025
NSQC Clearance Date	17/11/2022

Qualification Pack

SSC/N0512: Write, test and execute programs in HTML5 to develop web or mobile applications as per specifications

Description

This NOS is about code development using HTML5 along with other supporting languages such as JavaScript, CSS, etc.

Scope

The scope covers the following :

- Design Verification, Programming and Testing, Documentation

Elements and Performance Criteria

Design Verification

To be competent, the user/individual on the job must be able to:

PC1. Check the design specifications for the web or mobile app to be developed

PC2. Define the objectives and functionality of the program and align it to design specifications

Programming and Testing

To be competent, the user/individual on the job must be able to:

PC3. Develop and execute software code for various web/ mobile pages and applications using HTML5 in conjunction with CSS, JavaScript, jQuery, etc.

PC4. Implement coding best practices such as reusability, naming conventions, portability, etc.

PC5. Create appropriate test cases and test the web or mobile app

PC6. Debug the code and review the code with appropriate stakeholders

Documentation

To be competent, the user/individual on the job must be able to:

PC7. Maintain a record of test cases, software program, exception handling, reusable modules, etc

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. organizational policies, procedures and guidelines which relate to SDLC and programming

KU2. different data sources and how to access documents and information from data sources

KU3. organizational policies and procedures for documentation

KU4. who to involve while at various stages of SDLC

KU5. the range of standard templates and tools available and how to use them

KU6. overview of IR 4.0 technologies and their use cases

KU7. basic approaches in design thinking and agile methodologies

KU8. programming constructs and coding practices relevant to the programming language

Qualification Pack

- KU9.** tools, platforms and frameworks required for code development in the programming language
- KU10.** basic techniques to secure software code
- KU11.** tools for documentation, data visualization and analysis
- KU12.** fundamentals of requirement analysis

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** seek clarification and advice from appropriate people
- GS2.** listen effectively and orally communicate information accurately
- GS3.** work effectively in a customer facing environment
- GS4.** create accurate and well written documents/reports
- GS5.** understand the impact of business related decisions on organization
- GS6.** follow instructions, guidelines, procedures, rules and service level agreements
- GS7.** follow rule-based decision-making processes
- GS8.** plan and organize work to achieve targets and deadlines

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Design Verification</i>	8	14	-	-
PC1. Check the design specifications for the web or mobile app to be developed	4	6	-	-
PC2. Define the objectives and functionality of the program and align it to design specifications	4	8	-	-
<i>Programming and Testing</i>	25	47	-	-
PC3. Develop and execute software code for various web/ mobile pages and applications using HTML5 in conjunction with CSS, JavaScript, jQuery, etc.	10	20	-	-
PC4. Implement coding best practices such as reusability, naming conventions, portability, etc.	4	8	-	-
PC5. Create appropriate test cases and test the web or mobile app	6	10	-	-
PC6. Debug the code and review the code with appropriate stakeholders	5	9	-	-
<i>Documentation</i>	2	4	-	-
PC7. Maintain a record of test cases, software program, exception handling, reusable modules, etc	2	4	-	-
NOS Total	35	65	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N0512
NOS Name	Write, test and execute programs in HTML5 to develop web or mobile applications as per specifications
Sector	IT-ITes
Sub-Sector	IT Services
Occupation	Application Deployment
NSQF Level	4
Credits	9
Version	1.0
Last Reviewed Date	NA
Next Review Date	17/11/2025
NSQC Clearance Date	17/11/2022

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
5. In case of successfully passing only certain number of NOSs, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.
6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.
7. To pass a QF, a trainee should score a minimum aggregate of 70% across qualification

Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(**Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N0509.Verify the specifications and contribute to the design of software program	41	59	-	-	100	15
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	15
Total	61	89	-	-	150	30

Elective: 1 Java

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N0510.Develop, test and execute software programs as per specifications using Java	35	65	-	-	100	70
Total	35	65	-	-	100	70

Elective: 2 Python



Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N0511.Develop, test and execute software programs as per specifications using Python	35	65	-	-	100	70
Total	35	65	-	-	100	70

Elective: 3 HTML 5

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N0512.Write, test and execute programs in HTML5 to develop web or mobile applications as per specifications	35	65	-	-	100	70
Total	35	65	-	-	100	70

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.